

Project Title: BRCA1 Mutation Analysis in Breast Cancer Using cBioPortal & ClinVar

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Purpose: Self-directed learning project to understand cancer genomics databases and variant interpretation.

Tools Used:

- cBioPortal for Cancer Genomics
- ClinVar (NCBI)

Methods & Findings:

1. cBioPortal Analysis

- Selected TCGA Breast Invasive Carcinoma cohort (n=996 patients)
- Queried the BRCA1 gene
- **Result:** BRCA1 is altered in 44 out of 996 patients (**4%**)
- Mutation types observed: truncating, missense, amplification, deep deletion

2. ClinVar Validation

- Searched for the specific frameshift mutation c.68_69del (p.Glu23fs)
- **Result:** Classified as **Pathogenic** by ClinGen Expert Panel (June 2024)
- Associated condition: BRCA1-related cancer predisposition (hereditary breast and ovarian cancer)

Conclusion:

This project demonstrates ability to navigate real cancer genomics databases, interpret somatic mutation frequencies, and retrieve clinical significance of disease-associated variants. Skills are directly applicable to oncology research support and molecular diagnostics internships.